

VOXGEST CHAPTER 1 TO 3 APPROACH AND DECISION NOTES

Super-simple, defense-ready explanation of what was decided, why it was decided, where it belongs, when it happened, and how it connects to the system.

Main promise of this document

This document explains the decisions behind Chapter 1, Chapter 2, and Chapter 3 of VoxGest in very plain language. It is written so that even a very confused reader can follow the path: problem -> decision -> system feature -> paper section -> defense answer.

How to Use This PDF

Read this PDF as a map, not as a formal thesis chapter. It explains why the paper says what it says. It also explains why the system became static letters + dynamic words + NOTHING + token composer + Android + avatar instead of staying as a CNN-only alphabet recognizer.

- Use it before oral defense to understand the story behind Chapters 1 to 3.
- Use it with teammates so everyone explains VoxGest the same way.
- Use the decision cards when a panel asks, "Why did you choose this approach?"
- Use the chapter maps to check if the paper, system, and demo are connected.

One-Sentence Explanation

VoxGest in one sentence

VoxGest is an offline Android accessibility prototype that uses the phone camera to recognize selected one-hand signs, turn accepted signs into text and speech, and help the hearing person respond through speech-to-text and simple avatar or fingerspelling feedback.

The Whole Project Like a Simple Story

Part	Simple meaning
The problem	A Deaf or signing user may need to talk to a hearing person, but the hearing person may not know sign language and an interpreter may not be available.
The tool	A phone can sit between them like a small bridge.
The camera	The camera watches the signer.
The dot maker	MediaPipe turns the hand and upper body into landmark dots.
The model	The trained model reads the dots and guesses only the signs it was trained to know.
The guard	Gates check if the prediction is strong enough before allowing output.
The sentence builder	The token composer builds text only from accepted letters and words.
The voice	Text-to-speech can speak the accepted message out loud.
The reply path	Speech-to-text can turn the hearing person's reply into

	text.
The visual helper	The avatar or fingerspelling output gives the signing user a simple visual response.

Current Honest System Scope

The correct current scope must be repeated consistently in Chapter 1, Chapter 2, and Chapter 3. The system is not a full ASL grammar translator yet. It is a controlled sign-recognition and communication prototype.

Area	Current meaning
Static signs	A-Z, del, space, nothing. These support spelling and control actions.
Dynamic words	YES, NO, PLEASE, WATER, HELLO, HELP, STOP, DOCTOR, NAME, THANKYOU.
Negative class	NOTHING. It means no valid dynamic word should be displayed, spoken, or animated.
Sentence output	Built from confirmed accepted tokens, not from raw predictions.
Phrase intent	Explored as a future/optional path such as ASK_NAME, but phrase expansion is frozen during Recognition Hardening v1.
Android direction	On-device Android prototype using camera, landmarks, TensorFlow Lite, token composer, speech, and avatar response.

Chapter 1 to 3 Big Relationship Map

Part	Role in the whole project
Chapter 1	Explains the problem and the proposed solution. It answers why VoxGest exists.
Chapter 2	Explains the studies and technologies that support the solution. It answers why the chosen approach makes sense.
Chapter 3	Explains the method, system architecture, data, models, testing, and evaluation. It answers how VoxGest is built and checked.
System demo	Shows whether the paper claims are real. The demo must match the scope written in Chapter 1 to 3.

The Logic Path From Problem to Methodology

1. There is a communication barrier between sign language users and non-signers.
2. A phone-based accessibility tool can help because phones are common and portable.
3. The tool should work offline because daily public situations may not have stable internet.
4. The first model idea was CNN image recognition because CNNs are common for visual classification.
5. Live camera testing showed CNN-only recognition was fragile under real lighting, background, distance, and cropping conditions.
6. The team moved to landmarks because dots on the hand and body are more stable than whole images.
7. Static alphabet recognition uses 63 hand features because 21 hand landmarks times 3 coordinates equals 63.
8. Dynamic word recognition uses motion sequences because a moving sign cannot be understood from one frozen picture.
9. Dynamic signs use 30 frames x 162 features because motion needs time and some words depend on hand position near the face, chin, or body.
10. NOTHING was added because the model must learn when not to say anything.

11. Gates and stable-frame rules were added because raw model guesses must not immediately become output.
12. A token composer was added so only confirmed accepted predictions build the sentence strip.
13. Speech-to-text and avatar/fingerspelling were added because real communication must go both ways.
14. Phrase intents were explored, but phrase expansion is frozen because the current priority is to make the alphabet, 10 words, and NOTHING reliable first.

Chapter 1 Decisions: What the Paper Must Say and Why

Chapter 1 is the promise of the project. It must not promise more than the system can show. It must explain the real problem, the current scope, the objectives, and the limitations in a way that matches the actual system progress.

Chapter 1 part	Decision behind it
Background	Talk about communication barriers for Deaf, mute, speech-impaired, and hard-of-hearing users. This explains why the project exists.
Project context	Describe VoxGest as an offline Android accessibility prototype using landmarks, static recognition, dynamic word recognition, token composition, speech, and avatar response.
Purpose	Say that the app helps both sides communicate: sign user to hearing user, and hearing user back to sign user.
Statement of the problem	Ask questions that match the actual modules: Android app, landmarks, static letters, dynamic words, NOTHING, token composer, speech/avatar, and ISO evaluation.
Objectives	Turn the problem questions into measurable build goals.
Scope and limitation	Be honest: selected signs only, controlled vocabulary, no full ASL grammar yet, phrase intent explored but not the current priority.

Chapter 2 Decisions: Why These RRL Topics Belong

Chapter 2 must not be random. Every related literature topic must support a real part of VoxGest. If a study does not support the system, it weakens the paper.

RRL category	Why it belongs
Sign language recognition and accessibility	Supports the problem and shows why assistive sign recognition matters.
CNN/static gesture recognition	Explains the first approach and why image recognition was a reasonable starting point.
MediaPipe and landmarks	Supports the current feature-extraction decision.
Dynamic recognition using LSTM/TCN	Supports moving word recognition because dynamic signs happen over time.
Negative class and recognition hardening	Supports NOTHING, gates, false-positive control, and no-output handling.
Android edge deployment	Supports offline on-device inference and TensorFlow Lite deployment.
Speech-to-text and avatar feedback	Supports bidirectional communication instead of one-way sign-to-speech only.
ISO/IEC 25010	Supports the evaluation method for functional suitability, performance efficiency, reliability, and usability.

Chapter 3 Decisions: Why the Methodology Is Built This Way

Chapter 3 is the build map. It should explain what was done, why it was done, and how it was implemented. It should make the system defensible even if some parts are still prototypes.

Methodology part	Reason it belongs
Developmental research	The project creates and improves a working technology artifact.
Agile RAD	The team needed short cycles of building, live testing, fixing, and documenting.
System architecture	Shows the full path: camera -> landmarks -> models -> gates -> token composer -> text/speech/avatar.
Dataset development	Explains image data, webcam/manual data, word sequences, NOTHING samples, and calibration needs.
Feature engineering	Explains 63 static features and 30x162 dynamic sequences.
Model development	Shows the evolution from CNN to hybrid attempt to landmark model to LSTM/TCN dynamic models.
Recognition hardening	Explains gates, stability, cooldown, NOTHING, live logs, and why expansion is frozen.
Android development	Explains how the Python model contract becomes Android TFLite runtime behavior.
Avatar and speech module	Explains the reverse communication path and visual feedback.
Testing and evaluation	Explains static testing, word testing, negative testing, Android readiness, and ISO/IEC 25010 evaluation.

Detailed Decision Cards

Each decision below answers what, why, where, when, how, and relation. This is the easiest section to use for oral defense practice.

Decision 1: Build a phone-based accessibility app

Super simple meaning

Use one phone as a small bridge between a signing user and a hearing person.

Question	Answer in plain language
WHAT	Develop VoxGest as a mobile communication helper.
WHY	Phones are common, portable, and realistic for hospitals, schools, offices, counters, and homes.
WHERE	Chapter 1 background, project context, system scope, Android app.
WHEN	Started at project planning and stayed as the main product direction.
HOW	Use camera input, local recognition, text/speech output, speech-to-text, and avatar/fingerspelling.
RELATION	This decision defines the whole system. Every model and interface feature exists to support communication.
CHAPTER LINK	Chapter 1 explains the need; Chapter 3 explains how the phone app is built.

Decision 2: Make the system bidirectional

Super simple meaning

A real bridge must work both ways.

Question	Answer in plain language
WHAT	Support sign-user output and hearing-user response.
WHY	One-way sign-to-speech is incomplete because the hearing person still needs to reply accessibly.

WHERE	Chapter 1 purpose/objectives, Chapter 3 avatar and speech module.
WHEN	Scope was revised after the team realized sign-to-speech alone was not enough.
HOW	Add speech-to-text and avatar/fingerspelling response after accepted sign output.
RELATION	This makes VoxGest a conversation helper, not just a sign detector.
CHAPTER LINK	Chapter 1 must say bidirectional; Chapter 2 must support STT/avatar; Chapter 3 must include the module.

Decision 3: Use Android as the target platform

Super simple meaning

Python is the kitchen; Android is where the user eats the finished meal.

Question	Answer in plain language
WHAT	Prepare the app for Android phones.
WHY	Android supports mobile camera, local AI runtime, speech tools, and accessibility-focused deployment.
WHERE	Android handoff, Chapter 1 scope, Chapter 3 application development.
WHEN	Clarified during mobile handoff and current defense-ready system scope.
HOW	Use Kotlin, CameraX, MediaPipe/Holistic or equivalent landmarks, and TensorFlow Lite.
RELATION	The paper must describe Android as the main deployment direction, not a vague platform idea.
CHAPTER LINK	Chapter 1 says where it runs; Chapter 3 explains the Android method.

Decision 4: Keep recognition offline and on-device

Super simple meaning

The phone should think for itself instead of always asking the internet.

Question	Answer in plain language
WHAT	Run recognition locally using exported TFLite models.
WHY	Offline use supports privacy, lower delay, and use in weak-network environments.
WHERE	Chapter 1 context, Chapter 2 edge computing, Chapter 3 Android deployment.
WHEN	Established during scope refinement and Android developer handoff.
HOW	Train in Python, export to TFLite, and run inference on Android.
RELATION	This connects accessibility with practical real-world use.
CHAPTER LINK	Chapter 2 must include edge/mobile AI; Chapter 3 must explain TFLite runtime.

Decision 5: Start with CNN image recognition

Super simple meaning

At first, the AI looked at hand pictures like flashcards.

Question	Answer in plain language
WHAT	Train an image-based CNN on ASL alphabet images.
WHY	CNNs are a common first approach for visual recognition and were academically reasonable for static signs.
WHERE	Early model development and Chapter 3 model evolution.
WHEN	Started during the first ML phase using ASL alphabet image datasets.
HOW	Use image frames, convolution layers, and class labels for alphabet gestures.
RELATION	This was the first proof-of-concept, not the final strongest pipeline.
CHAPTER LINK	Chapter 2 can discuss CNN literature; Chapter 3 must explain it as an early approach.

Decision 6: Move away from CNN-only

Super simple meaning

The whole picture was too messy, so the system learned to focus on important dots.

Question	Answer in plain language
WHAT	Stop presenting CNN as the final main model.
WHY	Live testing showed CNN image recognition was sensitive to lighting, background, cropping, camera distance, and domain mismatch.
WHERE	Paper correction, model evolution, Chapter 3 methodology.
WHEN	After live webcam failures and hybrid testing.
HOW	Explain CNN as an attempted approach, then shift the final system to landmarks and temporal models.
RELATION	This keeps the paper honest and aligned with the current system.
CHAPTER LINK	Chapter 1 and 3 must not say CNN is the current main final method.

Decision 7: Use MediaPipe landmarks

Super simple meaning

Instead of staring at the whole room, the system follows dots on the hand and body.

Question	Answer in plain language
WHAT	Convert camera frames into hand and pose landmark coordinates.
WHY	Landmarks reduce background noise and make model inputs more stable across camera settings.
WHERE	Feature extraction in Chapter 3; landmark RRL in Chapter 2; Android runtime.
WHEN	Adopted after raw image and hybrid model limitations.
HOW	Use MediaPipe Hands/Holistic to extract x, y, z coordinates.
RELATION	Landmarks are the bridge between the camera image and model numbers.
CHAPTER LINK	Chapter 2 supports it; Chapter 3 explains how it works.

Decision 8: Use 63 static features for alphabet signs

Super simple meaning

One hand has 21 dots; each dot has x, y, z; $21 \times 3 = 63$.

Question	Answer in plain language
WHAT	Represent static hand signs using 63 normalized hand coordinates.
WHY	Static letters mostly depend on hand shape, so one-hand geometry is enough and efficient.
WHERE	Static alphabet model, Android runtime contract, methodology feature engineering.
WHEN	Used after landmark-only static model became the stronger direction.
HOW	Extract 21 hand landmarks, subtract wrist, scale by hand size, feed 63 floats into the model.
RELATION	This supports A-Z, del, space, and nothing.
CHAPTER LINK	Chapter 1 scope includes static signs; Chapter 3 explains the 63-feature method.

Decision 9: Use 30-frame sequences for dynamic words

Super simple meaning

A moving sign is like a mini-video, not one frozen photo.

Question	Answer in plain language
WHAT	Represent each dynamic word as a short sequence of frames.
WHY	Dynamic signs require motion order and timing, which one static picture cannot capture.
WHERE	Dynamic word model, Chapter 3 feature engineering.
WHEN	Added when the system expanded beyond alphabet recognition.
HOW	Collect or extract 30 frames for each sign attempt and feed the sequence to LSTM/TCN.
RELATION	This enables faster communication than spelling every word.
CHAPTER LINK	Chapter 1 objectives mention dynamic words; Chapter 3 explains the sequence method.

Decision 10: Use 162 features per dynamic frame

Super simple meaning

The AI needs to know if the hand is near the mouth, chin, chest, or body.

Question	Answer in plain language
WHAT	Use 99 pose features plus 63 dominant-hand features per frame.
WHY	Some signs cannot be distinguished by hand shape alone because body location matters.
WHERE	Dynamic feature engineering, Android model contract, methodology.
WHEN	Major upgrade during the holistic spatial-temporal pipeline stage.
HOW	Use pose landmarks and hand landmarks anchored relative to the body/face, producing 162 features per frame.
RELATION	This fixed the floating-hand problem where body location was missing.
CHAPTER LINK	Chapter 3 must explain why dynamic words use pose +

	hand, not hand-only.
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Decision 11: Solve the floating-hand problem

Super simple meaning

A hand near the mouth and a hand near the chest must not look the same.

Question	Answer in plain language
WHAT	Add body-relative spatial context to dynamic recognition.
WHY	Hand-only tracking removed location, causing confusion between signs that happen at different body areas.
WHERE	ML design decision, dynamic feature engineering, Chapter 3.
WHEN	Identified during live/dynamic word testing and addressed in the 162-feature holistic upgrade.
HOW	Anchor pose and hand features so the model can compare the hand position with the signer body/face.
RELATION	This explains why pose landmarks are not unnecessary extras; they solve a real recognition weakness.
CHAPTER LINK	Chapter 2 supports holistic/dynamic recognition; Chapter 3 explains the engineering fix.

Decision 12: Prioritize one dominant hand

Super simple meaning

The app follows one hand on purpose, not by accident.

Question	Answer in plain language
WHAT	Use a dominant-hand policy for recognition and data collection.
WHY	It supports users who sign with one hand by need or preference and keeps training/testing consistent.
WHERE	Chapter 1 accessibility rationale; Chapter 3 feature policy; Android handoff.
WHEN	Clarified after single-hand and two-hand data mismatch issues.
HOW	Set dominant hand as right, left, or controlled auto; use one selected hand in dynamic features.
RELATION	This gives VoxGest a clear accessibility position and avoids unstable hand switching.
CHAPTER LINK	Chapter 1 explains inclusivity; Chapter 3 explains implementation.

Decision 13: Mask non-dominant pose movement

Super simple meaning

The system should not secretly learn the wrong hand through body points.

Question	Answer in plain language
WHAT	Keep head, torso, and selected dominant arm while masking unrelated pose movement.
WHY	Old two-hand movement could leak into pose features even if only one hand vector was used.
WHERE	Single-hand feature policy, dynamic model, methodology.

WHEN	Added after single-hand data regression was found.
HOW	Apply a feature policy that preserves tensor shape but zeros non-dominant/unrelated pose landmarks.
RELATION	This keeps the model aligned with one-hand recognition without breaking Android compatibility.
CHAPTER LINK	Chapter 3 should explain this as a data/feature policy for reliability.

Decision 14: Add NOTHING as a negative class

Super simple meaning

Sometimes the correct answer is silence.

Question	Answer in plain language
WHAT	Train and handle NOTHING as no valid word/no-output.
WHY	Without NOTHING, a word classifier may force idle hands or partial gestures into one of the known words.
WHERE	Dynamic recognition, recognition hardening, Chapter 1 scope, Chapter 3 testing.
WHEN	Added during word-recognition fixes and Recognition Hardening v1.
HOW	Record idle hands, open hands, partial signs, transitions, aborted gestures, and hand entering/leaving as NOTHING.
RELATION	This prevents embarrassing false positives during demo and real use.
CHAPTER LINK	Chapter 1 scope includes NOTHING; Chapter 3 explains negative class testing.

Decision 15: Use gates before accepting predictions

Super simple meaning

The model guess must pass a security guard before it becomes output.

Question	Answer in plain language
WHAT	Check confidence, margin, motion, wrist path, hand presence, stability, and cooldown.
WHY	Raw AI predictions can be unstable; gates prevent weak guesses from becoming displayed/spoken words.
WHERE	Recognition hardening, live runtime, Chapter 3 post-processing.
WHEN	Added during live word reliability fixes.
HOW	Only accept predictions that pass metric thresholds and remain stable long enough.
RELATION	This makes the app safer because output is based on accepted tokens, not every softmax guess.
CHAPTER LINK	Chapter 3 must explain gates as part of methodology, not as random code details.

Decision 16: Use a token composer

Super simple meaning

Only clean bricks are allowed to build the sentence wall.

Question	Answer in plain language
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WHAT	Build the output sentence from confirmed accepted letters and word tokens.
WHY	It prevents rejected or unstable predictions from entering the message.
WHERE	Current demo system, Android runtime, Chapter 3 system architecture.
WHEN	Added in Recognition Hardening v1.
HOW	Accepted A-Z appends characters, space creates a word boundary, del deletes, accepted words append full tokens, NOTHING is ignored.
RELATION	This connects recognition output to readable conversation text.
CHAPTER LINK	Chapter 1 purpose mentions text output; Chapter 3 explains token composition.

Decision 17: Freeze phrase expansion during hardening

Super simple meaning

Do not build a second floor while the first floor is still shaking.

Question	Answer in plain language
WHAT	Disable phrase-intent expansion by default during the current hardening pass.
WHY	Phrase recognition can add confusion if static letters, 10 words, and NOTHING are not reliable yet.
WHERE	Progress log, current demo scope, paper limitation.
WHEN	Recognition Hardening v1 started on May 10, 2026.
HOW	Keep phrase files available but default phrase recognition off; focus on alphabet, demo10 words, and NOTHING.
RELATION	This protects the demo and makes the paper scope realistic.
CHAPTER LINK	Chapter 1 limitation and Chapter 3 current methodology must say phrase intent is explored/future, not the main current claim.

Decision 18: Explore endpoint phrase-intent separately

Super simple meaning

A full phrase should be judged only after the user finishes it.

Question	Answer in plain language
WHAT	Create a separate phrase-intent path for completed 60-frame gesture segments.
WHY	The first ASK_NAME test recognized too early because it was treated like a short word in a rolling window.
WHERE	Future phrase path, methodology note, progress documentation.
WHEN	Added on May 8, 2026 before Recognition Hardening v1 froze expansion.
HOW	Segment IDLE -> START_MOTION -> ACTIVE_MOTION -> END_HOLD, then classify once.
RELATION	This explains why phrase recognition must not use the same rolling 30-frame word model.
CHAPTER LINK	Chapter 3 can mention it as explored/future architecture; Chapter 1 should not overclaim full phrase translation.

Decision 19: Use LSTM and TCN for dynamic motion

Super simple meaning

Use models that can read a short movement over time.

Question	Answer in plain language
WHAT	Train temporal models for 30-frame dynamic word sequences.
WHY	LSTM and TCN can learn motion patterns; live behavior and speed may differ, so comparing them is useful.
WHERE	Dynamic model development, Chapter 2 RRL, Chapter 3 model development.
WHEN	Added after moving from static letters to moving words.
HOW	Feed 30x162 sequences into temporal models and compare validation plus live behavior.
RELATION	This supports dynamic word recognition better than static image classification.
CHAPTER LINK	Chapter 2 supports sequence models; Chapter 3 explains why dynamic signs need them.

Decision 20: Keep demo10 vocabulary fixed for now

Super simple meaning

Learn 10 useful words well before learning 50 badly.

Question	Answer in plain language
WHAT	Focus on YES, NO, PLEASE, WATER, HELLO, HELP, STOP, DOCTOR, NAME, THANKYOU, plus NOTHING.
WHY	Adding more words increases confusion and hides basic recognition problems.
WHERE	Scope, training protocol, current demo system, Chapter 1 limitation.
WHEN	Set as the defense-ready recognition hardening priority.
HOW	Train, audit, test, repair, and log only the current set before expanding.
RELATION	This gives the team a realistic 80-percent progress target.
CHAPTER LINK	Chapter 1 scope and Chapter 3 testing must match demo10.

Decision 21: Use live testing, not only validation accuracy

Super simple meaning

A test on saved files is not the same as a real camera test.

Question	Answer in plain language
WHAT	Evaluate model behavior in live webcam/Android-like conditions.
WHY	Validation accuracy may look good while live recognition fails due to camera angle, hand choice, lighting, and motion timing.
WHERE	Testing procedure, live logger, methodology.
WHEN	Used throughout dynamic recognition fixes and hardening.
HOW	Run diagnostic scripts, word-only mode, live test logger, and compare live logs.
RELATION	This makes retraining targeted instead of random.

CHAPTER LINK	Chapter 3 must include live testing because it is part of how the system is actually improved.
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Decision 22: Audit and repair datasets

Super simple meaning

Bad examples teach the model bad habits.

Question	Answer in plain language
WHAT	Check and clean the dataset before retraining.
WHY	Mixed hand modes, missing metadata, old samples, or contaminated recordings can lower recognition quality.
WHERE	Dataset development, training protocol, Chapter 3.
WHEN	Added after single-hand data regression and hardening priorities.
HOW	Use dataset audit reports, archive contaminated groups, re-record weak words and NOTHING samples.
RELATION	Good data is more important than randomly retraining many times.
CHAPTER LINK	Chapter 3 dataset development must explain data quality, not just dataset size.

Decision 23: Use speech output

Super simple meaning

The app talks for the sign user after a sign is confirmed.

Question	Answer in plain language
WHAT	Convert accepted text into spoken audio.
WHY	The hearing person may not read the screen quickly or may be across a counter.
WHERE	Purpose, system components, Chapter 3 output module.
WHEN	Included in the bidirectional communication design.
HOW	Use text-to-speech after token composition and accepted output.
RELATION	This completes the sign-user-to-hearing-user direction.
CHAPTER LINK	Chapter 1 purpose and Chapter 3 speech module must include it.

Decision 24: Use speech-to-text for replies

Super simple meaning

The hearing person can talk back into the same bridge.

Question	Answer in plain language
WHAT	Convert spoken response into text for the signing user.
WHY	If only the app speaks, communication still remains one-way.
WHERE	Bidirectional scope, Chapter 1 objectives, Chapter 3 speech module.
WHEN	Added during scope revision to bidirectional communication.
HOW	Use speech recognizer to turn hearing-user audio into text output.
RELATION	This supports the reverse communication path.

CHAPTER LINK	Chapter 1 must mention it; Chapter 3 must explain how it fits.
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Decision 25: Add lightweight avatar/fingerspelling

Super simple meaning

The app gives a visual reply instead of only text.

Question	Answer in plain language
WHAT	Show known-word animations or fingerspell unknown words.
WHY	Some signing users benefit from visual sign-like feedback; it also supports the bidirectional claim.
WHERE	Avatar pipeline, Android assets, Chapter 3 avatar module.
WHEN	Added as part of the reverse communication path and avatar prototype.
HOW	Known tokens load small JSON animations; unknown words are fingerspelled; NOTHING triggers no animation.
RELATION	This is a prototype visual response, not a complete ASL grammar avatar.
CHAPTER LINK	Chapter 1 describes visual response; Chapter 3 explains the lightweight avatar limitation.

Decision 26: Use ISO/IEC 25010 for evaluation

Super simple meaning

Judge the system like software, not only like an AI model.

Question	Answer in plain language
WHAT	Evaluate functional suitability, performance efficiency, reliability, and usability.
WHY	VoxGest must be useful, fast, stable, and understandable, not just mathematically accurate.
WHERE	Chapter 1 problem/objectives, Chapter 2 evaluation literature, Chapter 3 evaluation method.
WHEN	Used as the planned evaluation framework for professor review.
HOW	Prepare Likert-scale questionnaires for IT professionals and target/accessibility users.
RELATION	This connects system development to measurable capstone evaluation.
CHAPTER LINK	Chapter 3 must include respondents, instrument, and statistical treatment.

Decision 27: Use developmental research and Agile RAD

Super simple meaning

Build, test, fix, and repeat until the tool works better.

Question	Answer in plain language
WHAT	Use a development-focused research design and iterative methodology.
WHY	The project creates a working system and needs repeated live testing and revision.

WHERE	Chapter 3 research design and development methodology.
WHEN	Fits the whole development process from March to current hardening.
HOW	Brainstorm, design, develop, test, evaluate, and revise in short cycles.
RELATION	This explains why the system changed over time and why changes are valid research decisions.
CHAPTER LINK	Chapter 3 becomes the story of controlled iteration, not random trial and error.

Decision 28: Keep paper claims aligned with current system

Super simple meaning

The paper and demo must tell the same story.

Question	Answer in plain language
WHAT	Update Chapter 1-3 so claims match real progress.
WHY	If the paper says CNN-only but the system uses landmarks/LSTM/TCN, the panel will see inconsistency.
WHERE	All chapters, defense script, professor-review package.
WHEN	Current paper update stage.
HOW	Remove outdated CNN-only wording, add current scope, explain model evolution, and keep limitations honest.
RELATION	Alignment is what makes the paper oral-defense ready.
CHAPTER LINK	Chapter 1, 2, and 3 must connect: problem -> literature -> methodology -> actual system.

Chapter 1 - Defense-Ready Simple Notes

Chapter 1 must sound like the project is real, focused, and honest. It should not sound like a dream system that already translates all ASL. The safest wording is controlled selected sign recognition, bidirectional accessibility prototype, and current recognition hardening.

Background of the Study

- What it says: Deaf/signing users may struggle to communicate with non-signers when no interpreter is around.
- Why it says this: The system needs a real human problem, not just a technology experiment.
- How it connects: The problem justifies camera recognition, speech output, speech-to-text, and avatar feedback.

Project Context

- What it says: VoxGest is an Android-based offline assistive communication prototype.
- Why it says this: The current system has Android/TFLite handoff and local runtime goals.
- How it connects: The app uses landmarks and models locally, so it belongs under AI, computer vision, Android, and accessibility.

Purpose and Description

- What it says: The app recognizes selected signs, displays text, speaks output, accepts spoken replies, and gives visual/avatar response.
- Why it says this: The title says bidirectional, so the purpose must include both directions of conversation.
- How it connects: This section becomes the checklist for the system components in Chapter 3.

Statement of the Problem and Objectives

- What it says: The study asks how to design, implement, test, and evaluate the current system modules.
- Why it says this: Each problem question must lead to an objective and methodology step.

- How it connects: If there is an objective for dynamic words, Chapter 3 must explain dynamic word training and testing.

Scope and Limitation

- What it says: Selected signs only, current demo10 vocabulary, NOTHING, no full ASL grammar yet, phrase intent currently frozen/optional.
- Why it says this: Honest limitations protect the project from overclaiming.
- How it connects: A good limitation makes the project look realistic, not weak.

Chapter 2 - Defense-Ready Simple Notes

Chapter 2 must prove that the chosen technologies are not random. The literature should support each major decision in the system.

Sign language recognition and accessibility

- Supports the social problem.
- Explains why sign recognition tools matter.
- Connects to the target users and real situations.

CNN and deep learning

- Supports the first model attempt.
- Explains why CNN was reasonable at the start.
- Must not pretend CNN is still the final main approach.

MediaPipe and landmarks

- Supports the shift from raw images to coordinate features.
- Explains why landmarks are stable and portable.
- Connects directly to static 63 features and dynamic 162 features.

LSTM and TCN dynamic recognition

- Supports the need for temporal models.
- Explains why moving signs need sequences.
- Connects to 30-frame word recognition.

Negative class and recognition hardening

- Supports NOTHING and no-output logic.
- Explains why false positives matter.
- Connects to reliability and safety.

Android edge computing

- Supports offline phone-based inference.
- Explains why TensorFlow Lite and local processing matter.
- Connects to performance efficiency and privacy.

Bidirectional communication and avatar

- Supports speech-to-text and visual response.
- Explains why communication must go both directions.
- Connects to the title and purpose.

ISO/IEC 25010

- Supports the evaluation method.
- Explains why the system is measured using software quality characteristics.

- Connects to Chapter 3 respondents and questionnaire.

Chapter 3 - Defense-Ready Simple Notes

Chapter 3 must tell the panel exactly how the system is made. It should be written like a recipe: ingredients, steps, tests, and reason for every choice.

Chapter 3 section	Simple explanation
Research Design	Developmental research because the team is creating and improving a working application.
Development Methodology	Agile RAD because the team repeatedly builds, tests, fixes, and documents.
System Architecture	Camera -> landmarks -> models -> gates -> token composer -> text/speech/avatar.
Dataset Development	Static images, manual webcam samples, WLASL/selected videos, negative NOTHING samples, and calibration data.
Feature Engineering	63 features for static hand signs; 30x162 for dynamic signs; phrase 60x162 exists as future/optional architecture.
Model Development	CNN attempt, hybrid attempt, landmark static model, dynamic LSTM/TCN model, recognition hardening.
Recognition Hardening	Use NOTHING, gates, live logs, dataset audits, and frozen vocabulary to improve reliability.
Android Development	Mirror Python preprocessing on Android with CameraX, MediaPipe/Holistic, TFLite, and runtime labels.
Avatar and Speech Module	Text-to-speech, speech-to-text, known-word animations, unknown-word fingerspelling, no animation for NOTHING.
Testing Procedure	Static testing, dynamic testing, negative testing, Android readiness testing, and live behavior logging.
Evaluation Method	ISO/IEC 25010 using Functional Suitability, Performance Efficiency, Reliability, and Usability.

Common Panel Questions and Simple Answers

Panel question	Simple answer
Is VoxGest a full ASL translator?	No. It is a controlled sign-recognition accessibility prototype. It supports static letters, selected dynamic words, token-based output, and a visual response path. Full ASL grammar is future work.
Why did you not stay with CNN?	CNN was a reasonable first attempt, but live camera conditions made it fragile. Landmarks are more stable because they focus on hand/body geometry instead of the whole room.
Why use 63 features for letters?	A hand has 21 landmarks and each landmark has x, y, and z coordinates. That gives 63 features for static hand shape.
Why use 30x162 for dynamic words?	Dynamic words need motion over time. Each frame has pose and dominant-hand landmarks so the system knows both movement and body location.
Why include NOTHING?	NOTHING teaches the model when no valid word is happening. It reduces false output during idle hands, partial signs, and transitions.
Why one dominant hand?	The project focuses on one-hand accessibility and stable mobile recognition. This can support users who sign with one hand by need or preference.
Why freeze phrase recognition?	Phrase recognition was explored, but the current priority is to make the alphabet, 10 words, and NOTHING

	reliable before expanding.
Why Android?	Android phones are common and can run camera, landmarks, TensorFlow Lite, speech, and avatar functions locally.
Why ISO/IEC 25010?	The system must be evaluated as software: does it work, is it fast enough, is it reliable, and can users understand it?

What Must Never Be Said Incorrectly

Type	Statement
Wrong	VoxGest fully translates ASL sentences.
Correct	VoxGest recognizes selected signs and builds output from confirmed accepted tokens. Full ASL grammar is future work.
Wrong	CNN is the final main model.
Correct	CNN was an early approach. Current recognition uses landmarks and temporal models.
Wrong	NOTHING is a word users say.
Correct	NOTHING is a no-output negative class used to reduce false recognition.
Wrong	Phrase intent is the current main feature.
Correct	Phrase intent was explored, but phrase expansion is frozen during Recognition Hardening v1.
Wrong	The avatar is a full ASL avatar.
Correct	The avatar is a lightweight visual response prototype with known-word animations and fingerspelling fallback.

Final Chapter 1 to 3 Alignment Checklist

Checklist item	Status
Chapter 1 background explains the communication barrier.	Yes / No
Chapter 1 purpose says bidirectional communication.	Yes / No
Chapter 1 objectives match actual modules.	Yes / No
Chapter 1 limitation avoids full ASL translation overclaim.	Yes / No
Chapter 2 RRL supports landmarks, temporal models, edge Android, speech/avatar, and ISO evaluation.	Yes / No
Chapter 2 explains CNN as related/early approach, not final-only system.	Yes / No
Chapter 3 explains 63 static features.	Yes / No
Chapter 3 explains 30x162 dynamic features.	Yes / No
Chapter 3 explains NOTHING and gates.	Yes / No
Chapter 3 explains token composer.	Yes / No
Chapter 3 includes Android/TFLite handoff logic.	Yes / No
Chapter 3 includes avatar and speech response module.	Yes / No
Chapter 3 includes ISO/IEC 25010 evaluation plan.	Yes / No

Team Reading Path

Use this path when reading the paper as a team:

15. Read Chapter 1 and ask: Are we honest about what the system can currently do?
16. Read Chapter 2 and ask: Does every RRL topic support one actual system decision?
17. Read Chapter 3 and ask: Can someone follow the build process from camera input to output?

18. Compare the paper with the demo and ask: Does the demo prove the same scope written in the paper?
19. Practice the panel questions and make sure all team members give the same explanation.

Source Basis Used for This Companion PDF

This companion PDF was based on the current VoxGest project sources available in the workspace, including the current demo system notes, latest knowledge update, Android developer handoff, recognition hardening progress logs, ML engineer knowledge base, recording protocol, avatar pipeline, Chapter 2-3 research notes, the current Chapter 1-2 paper, the defense-ready Chapter 1-3 preview, and the existing simplified/contextual notes. The content is summarized in plain language and does not replace the formal thesis paper.

Source group	Used for
Current demo system / README_CURRENT	Scope: static alphabet, demo10 dynamic words, NOTHING, token composer, phrase off by default.
Android Developer Handoff	Android runtime contract: CameraX, MediaPipe/Holistic, TFLite, static [1,63], dynamic [1,30,162], post-processing.
Progress logs	Timeline: phrase-intent added, then Recognition Hardening v1 froze phrase expansion and prioritized reliability.
Latest Knowledge Update	Dominant-hand policy, single-hand pose policy, model contract, NOTHING, and 30x162 dynamic model.
ML Engineer Knowledge Base	Plain explanation of what VoxGest can and cannot do yet.
Avatar Pipeline	Known-word animations, fingerspelling fallback, NOTHING as no animation/no output.
Current paper and defense-ready preview	Chapter 1-3 content direction and corrected paper structure.

Final Simple Summary

The safest defense story

VoxGest started as a CNN alphabet idea, but real testing showed that a stronger approach was needed. The system moved to landmark-based static recognition and temporal dynamic recognition because those methods better match real camera use. The current goal is not full ASL translation. The current goal is a reliable offline Android accessibility prototype that recognizes selected signs, blocks weak outputs with NOTHING and gates, builds text only from accepted tokens, speaks confirmed output, and gives a simple visual reply through avatar or fingerspelling.